

Kenneth Oswin

Software Engineer

Python • Backend • Automation Engineering

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Portfolio • LinkedIn • GitHub • LeetCode

Professional Summary

Software Engineer with experience developing Python applications, backend services, AI-powered solutions, and reusable engineering frameworks. Skilled in REST APIs, CI/CD, developer tools, and automation using Python, Robot Framework, and Playwright. Passionate about applying AI and software engineering principles to build scalable and maintainable systems.

Professional Experience

Crestron

July 2025 – Present

Associate Software Development Engineer in Test

- Design and maintain Python-based engineering frameworks supporting multiple enterprise products, creating reusable libraries and infrastructure that accelerate software delivery across engineering teams.
- Developed an entire Robot Framework library from the ground up for an unreleased hardware product, implementing reusable abstractions over complex hardware interfaces while enabling scalable framework development.
- Designed backend API services and endpoints enabling hardware operations that previously required manual intervention, expanding platform capabilities beyond UI-driven workflows.
- Engineered CI/CD pipelines using Jenkins and Docker to execute automation suites continuously across multiple systems, improving execution reliability and reducing manual effort.
- Developed internal dashboards, execution trackers, reporting utilities, and engineering tools that improved framework visibility, monitoring, and execution management.
- Worked across three independent enterprise products, rapidly adapting framework architecture, reusable libraries, and engineering workflows to different software and hardware ecosystems.
- Collaborated closely with software developers during feature development, improving engineering feedback loops through scalable framework design and infrastructure enhancements.
- Built reusable Python modules, Robot Framework keywords, and shared engineering utilities that significantly reduced duplication while improving long-term maintainability.
- Helped standardize CI/CD environments across multiple engineering systems by configuring Jenkins instances, Docker environments, and execution infrastructure.
- Mentored two interns in Python, Robot Framework, framework architecture, and engineering best practices, both of whom successfully transitioned into full-time engineering roles.
- Contributed to expanding automation knowledge among functional engineers by introducing reusable development practices, framework design concepts, and engineering workflows.

Crestron

January 2025 – July 2025

Software Development Engineering in Test Intern

- Developed Python-based automation scripts and reusable Robot Framework components supporting products under active development.
- Integrated framework execution with Jenkins pipelines and Jira workflows, improving continuous validation and engineering productivity.

- Collaborated with software engineers to improve framework coverage, identify defects, and enhance overall product quality.
- Contributed reusable framework modules that became the foundation for future engineering infrastructure.

Projects

KenBot OS — Autonomous Multi-Agent AI Platform

2025 – Present

- Architected and developed a modular AI platform that autonomously manages content generation, scheduling, messaging, analytics, and cross-platform workflows across WhatsApp, Reddit, YouTube, X (Twitter), and web services.
- Designed a service-oriented backend using Python and Flask, integrating REST APIs, Playwright, APScheduler, SQLite, and Claude AI to orchestrate intelligent background services and autonomous workflows.
- Implemented persistent conversational memory, adaptive user profiling, contextual learning, and relationship-aware interactions to deliver long-term personalized AI experiences.
- Built an AI-powered content pipeline capable of generating social media posts, YouTube Shorts, trend-aware responses, blogs, and automated publishing schedules through intelligent orchestration.
- Developed reusable modules for AI orchestration, prompt management, browser automation, notifications, logging, analytics, and service communication, enabling scalable expansion into distributed multi-agent systems.
- Integrated external APIs, browser automation, scheduled workers, and monitoring services into a unified architecture emphasizing modularity, maintainability, and extensibility.

Veriq — AI Software Engineering Platform

2026 – Present

- Building an AI-powered software engineering platform that converts natural language requirements into production-ready software workflows spanning UI, API, and system layers.
- Architected a scalable monorepo with dedicated modules for backend services, React frontend, AI agents, analytics, integrations, deployment tooling, browser extensions, SDKs, CLI utilities, and developer documentation.
- Designed backend services using Python and Clean Architecture principles with clear separation of domain, infrastructure, and application layers for long-term maintainability and scalability.
- Developed REST APIs, health monitoring services, version management endpoints, and Docker-based development environments supporting reproducible local development and CI/CD pipelines.
- Designed the platform for future support of autonomous AI agents, self-healing workflows, intelligent execution planning, Jenkins plugins, GitHub Apps, language SDKs, and enterprise integrations.
- Established engineering standards through architecture documentation, API specifications, deployment guides, developer onboarding, security documentation, and roadmap planning.

Technical Skills

Programming Languages

Python | Java | JavaScript | TypeScript | C

Software Engineering

FastAPI | Flask | Django | REST APIs | Robot Framework | Playwright | Selenium

AI & Intelligent Systems

Claude API | OpenAI API | Gemini API | Prompt Engineering | OpenCV | scikit-learn | Pandas

Infrastructure & DevOps

Docker | Jenkins | Git | GitHub Actions | CI/CD Pipelines

Databases

SQLite | MySQL | MongoDB

Developer Tools

VS Code | Postman | Jira | GitHub

Leadership & Extracurricular Activities

- Served as **Creative Head** of **Anveshana**, the college magazine, leading branding, visual design, publication layouts, and creative direction.
- Led the creative and design team for **FantomCode**, the college hackathon, overseeing event branding, promotional material, and participant-facing assets.
- Served as **Creative Head** for **RobotFiesta**, directing event identity, digital media, and promotional campaigns.
- Progressed from Organizing Committee member in the Technical and Design teams to leading creative initiatives across multiple technical events.

Education

RV Institute of Technology and Management Bachelor of Engineering Information Science and Engineering (VTU) CGPA: 8.36/10	2021 – 2025
St. Joseph’s Pre-University College Pre-University Education (State Board) Percentage: 94%	2019 – 2021
Bishop Cotton Boys’ School ICSE Percentage: 94.6%	2007 – 2019